



AI-Powered Riverbank Erosion Risk Prediction and Early Warning System for Bangladesh

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1. Introduction

1.1 Background Overview

Application domain. Environmental monitoring and disaster early warning. Bangladesh is a delta country where large rivers such as the Jamuna, Padma and Meghna constantly shift their banks, and riverbank erosion destroys homes, farmland and embankments every year. The problem is recurring and sudden: at least 386 houses were washed away along the Teesta in five northern districts within a single week in September 2026, and more than 300 houses were lost in a month in Kurigram's Nageshwari upazila. Families often lose their belongings because collapse happens within minutes, and protection measures such as geo-bags are applied reactively and fail to stop the erosion without permanent riverbank protection. This makes timely, local, forward-looking risk signals valuable for vulnerable communities.

Current practices. Monitoring today relies mainly on:

- (a) river water-level gauge stations,
- (b) manual field inspection and local observation,
- (c) historical erosion records and maps, and
- (d) satellite imagery and GIS analysis.

These give valuable regional information, but they report what the river is doing or has already done. None of them gives a small community a localized, automatic, forward-looking signal.

Use of AI in this domain. Machine Learning is already used in flood forecasting, hydrology and remote-sensing studies. These systems usually depend on large datasets, specialized infrastructure or large-scale deployment, and they are rarely available at the level of a single vulnerable village bank. Low-cost, locally deployed AI for erosion risk is largely absent.

1.2 Problem Statement

Riverbank erosion is caused by a combination of factors: water level and its rate of change, rainfall, soil moisture and bank instability. Today, assessing the danger at a specific bank depends on people visiting and judging it by eye. This has four limitations:

- Continuous watching of every vulnerable location is not possible by manual observation.

- Reading several indicators together and judging how they combine is subjective and depends on individual experience.
- Existing systems mostly describe the present situation and do not estimate the chance of erosion in the coming hours.
- Warnings reach communities late, or not at all.

Who is affected: families, farmers and landowners living near eroding banks, and the local authorities responsible for them.

Why software is needed: measurements must be collected continuously, stored, processed and shown to non-experts in a simple form, which is not practical on paper.

Why AI is needed: today a human expert manually compares current water level, rainfall and bank condition against what they remember from past erosion events. The AI replaces this manual judgment by learning the relationship between those indicators and past erosion events from data, then applying it automatically to every new reading to give a probability of erosion within a future time window.

The project is scoped as a student prototype that predicts risk, not exact erosion location or land loss, and it makes no accuracy claims before real evaluation.

1.3 Objectives

General Objective: To develop a low-cost IoT and AI-based prototype that continuously monitors riverbank conditions and warns of rising erosion risk through a web dashboard and a local alert unit.

AI Objective: To train and evaluate a supervised Machine Learning model that takes environmental and bank-condition features as input and outputs the probability of erosion risk within a future time window, and to compare candidate algorithms to select the most suitable one.

1.4 Scope

In-scope	Out-of-scope
ESP32-based collection of water level, rainfall, soil moisture, bank tilt/movement, vibration, temperature and humidity (where feasible)	Deployment along an entire river or multiple sites; the prototype targets one monitoring station or a model setup

In-scope	Out-of-scope
Wireless transmission to a server and database storage	Building bank-protection structures or controlling river flow
Historical data integration, preprocessing and feature engineering	Predicting the exact erosion location or amount of land loss
ML-based erosion-risk prediction and Low/Medium/High/Critical classification	Replacing official disaster-management or hydrological authorities
Web dashboard with live readings, trends and risk level	Geological surveys, permanent commercial deployment, mobile app and SMS
Local warning using LEDs and a buzzer	Deep learning on satellite imagery
Model comparison, evaluation and feature-importance analysis	-

The AI will	The AI will not
Learn patterns from historical and sensor data; output an erosion probability and risk level for a future time window; show which inputs mattered most; support early-warning decisions	Guarantee that erosion will or will not happen; stop erosion automatically; replace expert hydrological or geological assessment; claim accuracy beyond what is measured on test data

Boundaries: one prototype station, a fixed set of sensors, decision support only. Any new sensor or feature outside this list is treated as future work.

1.5 Stakeholders

Stakeholder	Role	Interest in the system
Riverbank residents, farmers and landowners	End users	Receive early warning to move people, livestock and crops in time
Local authorities	decision makers	Use risk information for local planning and response

Stakeholder	Role	Interest in the system
Disaster-management and water-resource organizations	Potential clients and regulatory bodies	Use the output only as supplementary information; the system must not conflict with official guidance
System administrator	Operator	Maintains devices, server, database and model

1.6 Proposed Solution

The solution is an end-to-end pipeline. An ESP32 node reads the sensors every few minutes and sends the data over Wi-Fi to a backend API, which stores it in a database. A preprocessing step cleans the data and builds features such as water-level change and accumulated rainfall. The **trained ML model** converts these features into an erosion probability, which is mapped to a risk level. The level is shown on the dashboard and sent back to the ESP32 to drive LEDs and a buzzer. To avoid false alarms from a single noisy reading, a warning is raised only when high risk persists over several consecutive predictions.

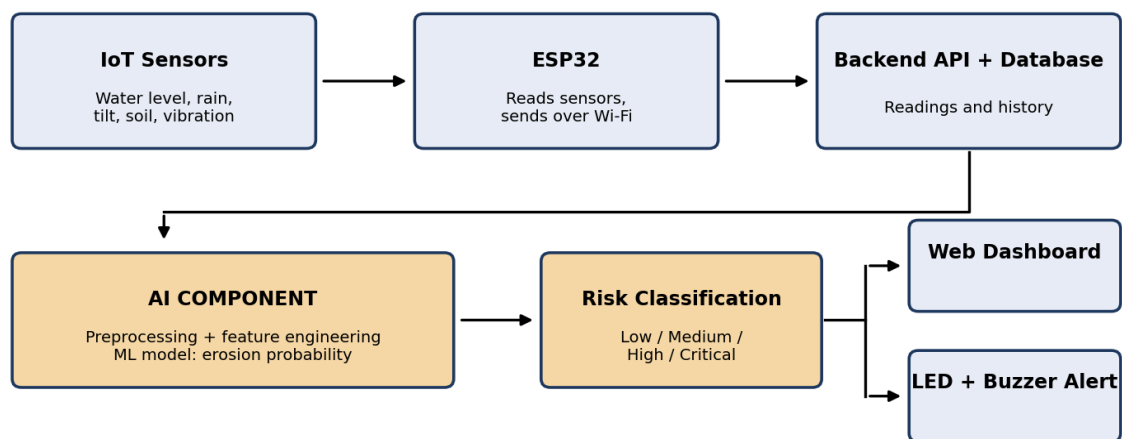


Figure 1. System overview. Shaded boxes mark the AI component.

1.7 AI Features

Existing AI features and limitations

- **Flood and water-level forecasting models:** predict river level, not bank erosion, and run at regional scale.

- **Satellite/remote-sensing erosion analysis:** shows past bank-line changes with delays of days or weeks, and cannot warn within hours.
- **Large hydrological models:** need big datasets and specialist expertise, and are not available for a single village bank.

Proposed AI features

Feature	Input	Output	Manual work it replaces
F1. Erosion-risk prediction	Water level and its change, accumulated rainfall, soil moisture, bank tilt, vibration, temperature, humidity	Erosion probability for the next time window	An expert mentally combining many indicators to judge danger
F2. Risk classification	Predicted probability plus recent predictions	Low / Medium / High / Critical	Deciding by eye when a situation is serious enough to warn
F3. Feature-importance analysis	Trained model and test data	Ranked list of influential factors	Guessing which factors drive erosion at this site

AI techniques

Supervised Machine Learning on structured, time-based data. Candidate models are Logistic Regression (baseline), Random Forest, XGBoost and Support Vector Machine, compared on a time-ordered test set. Feature importance uses Random Forest/XGBoost importance and optionally SHAP. Deep models such as LSTM are not planned unless enough sequential data is collected.

Why AI is essential

If the AI is removed, the system can only display raw sensor values and fixed alarm thresholds. It can no longer do its main job: converting many changing measurements into a forward-looking erosion probability learned from past events. The prediction, classification and explanation features are all produced by the ML model, so the AI is the core of the product and not an add-on.

2. System Requirements

2.1 Functional Requirements

ID	Requirement
FR-01	The system shall read water level, rainfall, soil moisture, bank tilt, vibration, temperature and humidity through an ESP32.
FR-02	The system shall send each reading to the server over Wi-Fi with a timestamp and station ID.
FR-03	The system shall store all readings and historical data in a database.
FR-04	The system shall detect missing or invalid readings and handle them without crashing.
FR-05	The system shall compute derived features, including water-level change and accumulated rainfall.
FR-06	The system shall show live readings and historical trend graphs on a web dashboard.
FR-07	The system shall show the device/communication status, including the time of the last received reading.
FR-08	The system shall activate an LED and buzzer when the risk level is High or Critical.
FR-09	The system shall require login for dashboard access and an API key for sensor data submission.

2.2 AI Requirements

ID	Requirement
AI-01	The system shall train the model on historical labeled data and be retrainable on new data.
AI-02	The system shall use a chronological train/test split to prevent future-data leakage.
AI-03	The system shall produce an erosion probability (0 to 100%) for each new observation, with its prediction time window.
AI-04	The system shall map the probability to Low, Medium, High or Critical using thresholds chosen on validation data.

ID	Requirement
AI-05	The system shall raise a warning only when high risk persists across a set number of consecutive predictions.
AI-06	The team shall evaluate at least three algorithms using Accuracy, Precision, Recall, F1-score and ROC-AUC, and report the results on the test set.
AI-07	The system shall display the most influential input features for the trained model.
AI-08	The dashboard shall present predictions as risk estimates with a note that they are not guaranteed.

2.3 Non-Functional Requirements

- **Performance:** a new reading shall appear on the dashboard together with its prediction within one minute under normal network conditions.
- **Reliability:** the system shall keep running and show a status warning when a sensor or the network fails.
- **Usability:** a non-expert shall be able to read the current risk level from the dashboard home page.
- **Maintainability:** IoT, backend, database, ML and dashboard modules shall be separate.
- **Scalability:** the data model shall support adding more stations later.

3. Tools and Technologies

Category	Technologies
Hardware	ESP32; JSN-SR04T waterproof ultrasonic sensor (water level); MPU6050 (bank tilt/vibration); soil-moisture sensor; DHT22 (temperature/humidity); rain sensor or tipping-bucket gauge; LEDs, buzzer; power bank; enclosure
Programming languages	Python, C/C++ (Arduino framework), JavaScript, SQL, HTML/CSS
AI / data libraries	Pandas, NumPy, Scikit-learn, Google Colab

Category	Technologies
AI models	Logistic Regression, Random Forest, XGBoost, SVM (own models trained on project data; no external LLM API)
Backend	FastAPI ; model served as a saved file loaded by the backend
Database	MySQL
Frontend	HTML, CSS, JavaScript, Chart.js ; OpenStreetMap for the station map
Development tools	Arduino IDE or PlatformIO, VS Code, Git/GitHub

4. Project Timeline and Work Plan

The project runs for 6 weeks. Team size is 3;

Week	Milestone	Main tasks	Deliverable
1	M1: Planning	Requirements, literature and data-source study, define labels and prediction window	Requirement and data plan
2	M2: Data and IoT	Collect/clean historical data; assemble ESP32 and sensors; test readings	Initial dataset, IoT prototype
3	M3: Backend	Database schema, REST API, feature engineering	Working backend, processed dataset
4	M4: ML model	Train and compare models; time-based evaluation; feature importance	Trained model and evaluation results
5	M5: Integration	Dashboard; connect ESP32, backend and model; alert unit	Integrated system
6	M6: Testing and finalization	System and model testing, bug fixes, report, demo	Final prototype, report, presentation

5. Optional Sections

5.1 Existing System

River-level gauge stations, field inspection, historical records and satellite/GIS analysis are the current tools. They are important but regional, manual or delayed. This project adds a localized, automatic and predictive layer and does not replace them.

5.2 System Architecture

The architecture is layered:

- (1) **sensing layer** (sensors + ESP32),
- (2) **communication layer** (Wi-Fi, REST API),
- (3) **data layer** (database),
- (4) **intelligence layer** (preprocessing, ML model, risk classification), 5) **presentation layer** (dashboard and LED/buzzer alert).

Figure 1 shows the data flow.

5.3 Future Work

- Multiple stations and LoRa/LoRaWAN long-range communication
- Better flow-velocity and rainfall sensors
- Satellite-image-based erosion detection and GIS mapping
- Deep learning when enough time-series data exists
- Mobile app and SMS alerts; integration with official platforms
- Field validation across different rivers and river-specific models

Risks and Assumptions

Labeled data is the main risk. A supervised model needs examples of past erosion and non-erosion periods linked to environmental conditions. The team will first confirm what labeled historical data can be obtained (for example public hydrological records and erosion reports). If real labels are limited, the prototype will be evaluated on the available data and any simulated or proxy

data will be clearly marked as such. No accuracy figures are claimed in this proposal; they will be reported only after training and testing.